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Introduction of 3-tier Architecture in DBMS.

What is 3 Tier Architecture in DBMS ?

- In DBMS, the 3 tier architecture is a client-server architecture that seperates application processing and data management into three distinct tiers or layers.
- Used in modern web applications and enterprise system.
- It offers scalability, flexibility and security.

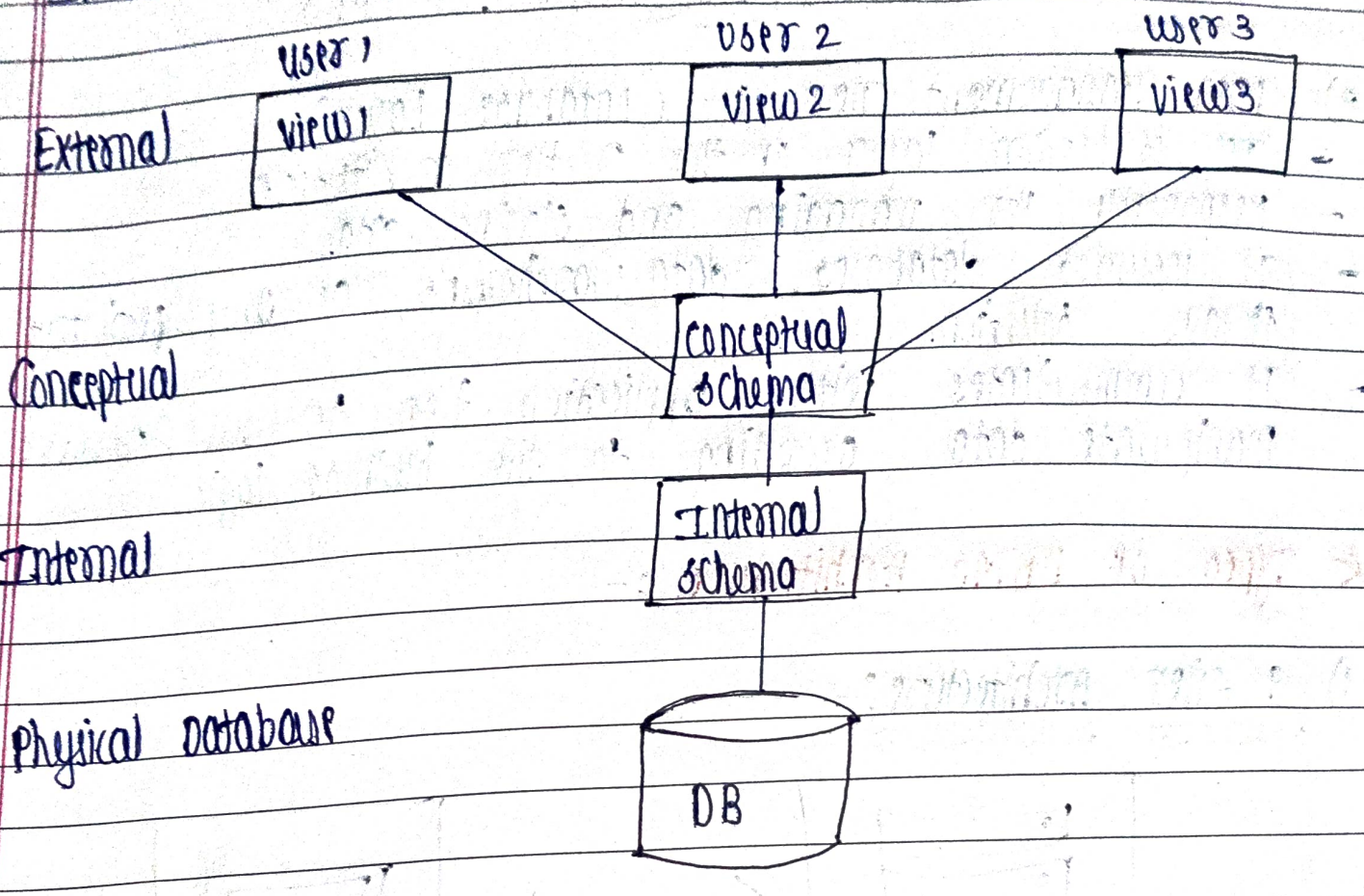


fig :- 3 tier Architecture.

Components of 3 tier Architecture :-

1) Presentation Tier :- (User Interface Layer) :-
- It is user interface or client layer of application.
- Responsible for presenting data to user and taking input.
- Processes user request and displays relevant information.

Ex :- Web browser, mobile app or desktop application.

2) Application Tier :- (Business logic Layer)
- It is an middle layer of 3-tier architecture.
- acts as intermediary between presentation & data management tier
- Responsible for processing and managing business logic of application
- It communicates with presentation tier to receive user input and communicates with data management tier to retrieve or store data.
- Ex:- Application servers, web servers, or API's.

3) Data Management Tier :- (Database Layer)
- It is bottom layer of the 3-tier architecture.
- Responsible for managing and storing data.
- It includes databases, data warehouses, or any persistent data storage solution.
- It communicates with application tier to save, retrieve or manipulate data according to the business logic.

★ Types of DBMS Architecture :-

1) 1-Tier Architecture :-

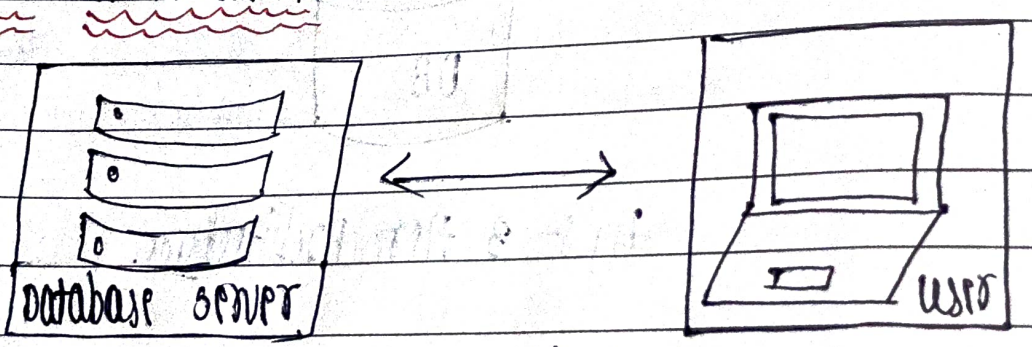
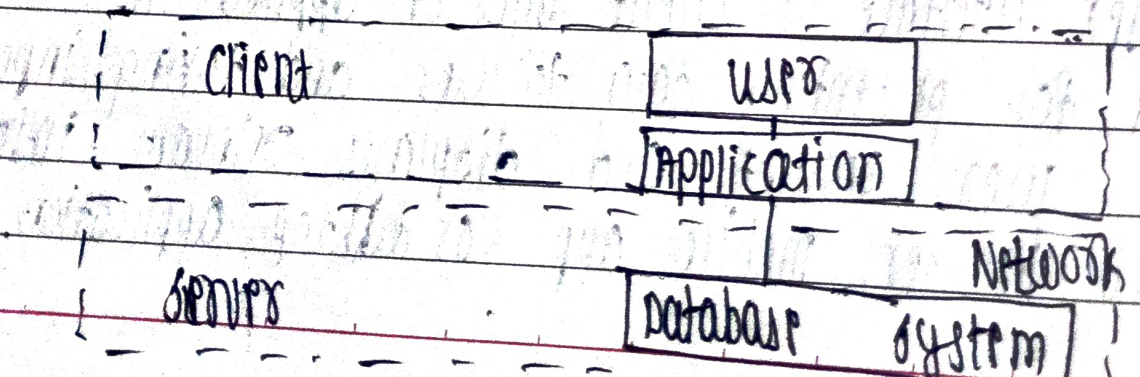
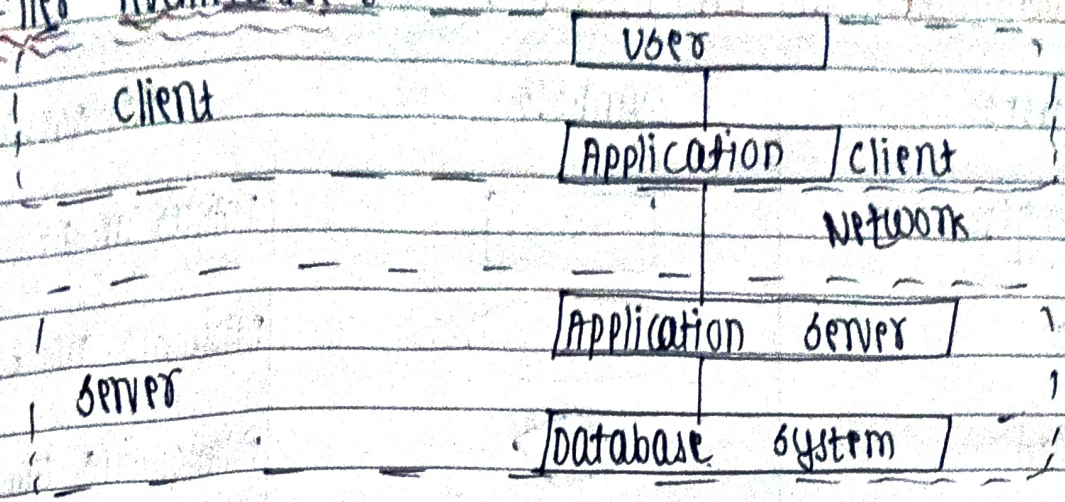


fig :- 1-tier Architecture.

2) Two-Tier Architecture :-



3) 3-Tier Architecture :-



* Introduction of ER Model :-
- It is a conceptual model for designing a database.
- It represents structure of a database including entities, attributes and relationship between them.

(i) Entity :-
An object that is stored as data.
Ex:- Student, course or company.

(ii) Attribute :-
Properties that describes an entity.
Ex:- StudentID, CourseName or EmployeeEmail.

(iii) Relationship :-
A connection between entities.
Ex:- student enrolls in a course.

